

Pair–Encircling Loops as Quarks

The Poincaré–dual edge basis, flavor from the diquark loop,
and why a two–quark loop is a diquark, not a gluon

Tessera theory notes (companion to `spin.readout.tex`; part of issue #410)

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Abstract

This is a research–direction companion to `spin.readout.tex` (same directory). It takes seriously a single proposal: make the *loops that encircle pairs* of color holes the primary objects — “the quarks” — rather than the holes themselves, and ask what flavor and what gauge content they carry. Three things follow, two of them theorems and one a reframing. (i) A pair–encircling cycle is, by homological additivity together with the carried singlet constraint $w_A + w_B + w_C = 0$, the Poincaré *dual* of the *complementary* hole: $[\text{loop around } i, j] = -[\text{hole } k]$. So “loops as quarks” is not a new set of three objects; it is the *edge basis* of the very same three quarks, carrying the identical $[1, \omega, \omega^2]$ color singlet. (ii) Flavor read on the loops gives uud with the *diquark loop* carrying d — and this is exactly the dual of the d –hole (the quark *outside* the diquark), so it matches the two–step diquark build of #489 with no inversion and no fiat; the loop combinatorics reproduce the proton’s mixed–symmetry skeleton under the diquark’s internal exchange. (iii) Group theory settles the “gluon” alternative: a loop around two *quark* holes lives in $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ — a *diquark* ($\bar{\mathbf{3}}$), *not* a gluon ($\mathbf{3} \otimes \mathbf{3} = \mathbf{8} \oplus \mathbf{1}$); a genuine gluon loop must enclose a quark *and an antiquark*, which exist across the proton/antiproton legs of the pair–creation build. **Status: a lens, not a validated result.** The *spin* readout this picture motivates reduces to the closed–loop holonomy and pairwise correlator of `cartan.weyl.gluon.tex`; it sharpens *what* the loops are and *which* flavor/color they carry, and it does *not* bypass the hard fiber $\leftrightarrow$ cells kernel those notes name.

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1 Scope: what this companion adds, and what it does not

Three sibling documents in this directory have already settled the surrounding problem, and we build directly on them rather than restating them. We write [SR §*n*] and [CW §*n*] for sections of, respectively, `spin_readout.tex` (the study document: the proton’s $J^2 = \frac{3}{4}$ is an entangled mixed-symmetry quantity, and a per-hole product read provably floors at the mixture $\sim \frac{9}{4}$; it names the cure, the total-space operator $\hat{J}^2 = \frac{3}{4} + \sum_{i<j} \hat{P}_{ij}$ keeping the cross-hole terms) and `cartan_weyl_gluon.tex` (the Cartan/Weyl lens: the color group’s Weyl S_3 is the swap S_3 ; the missing object is the abelian A factor, coordinatized by the connected pairwise correlator C_{ij} ; the entangling exponential acts like the one-gluon-exchange hyperfine interaction; and the fiber $\leftrightarrow$ cells lift with the closed-loop holonomy “escape hatch” — the consolidated home of the retired total-space note).

The one new move. All three siblings keep the three *holes* as the objects and add pairwise *operators* (swaps P_{ij} , correlators C_{ij}) on top. This note instead takes the user’s proposal literally: promote the three *loops* $\{\gamma_{AB}, \gamma_{AC}, \gamma_{BC}\}$ encircling the three pairs to the primary degrees of freedom and ask what they are. The payoff is not a new spin mechanism — §3 shows the spin readout reduces to [CW §6] — but a sharper account of (a) the *topology* of the reframe (§2: it is the Poincaré-dual edge basis), (b) *flavor* (§4: the diquark loop is d , dual to the d -hole), and (c) the *gluon* question (§5: rep theory makes the loop a diquark).

Status discipline. §2 and §5 are theorems (homological additivity; $\mathbf{3} \otimes \mathbf{3}$ Clebsch–Gordan). §4 is a consistent *labeling* that reproduces the proton’s symmetry skeleton — necessary, not sufficient, for $\frac{3}{4}$. The reframe does not, by itself, produce a number; §6 states exactly where it reduces to the already-open kernel.

2 The pair-encircling cycle is the dual of the complementary hole

Setup. The host is a closed combinatorial S^4 with three color holes A, B, C (three disjoint removed top cells, $b_3 = n - 1 = 2$ independent classes for $n = 3$ holes) [SR §5–7]. The carried representative is a single harmonic 3-cochain $\psi = \text{EigenstateSynthesis}(\text{st}, 3). \text{carriedRepresentative}$, and each hole h is detected by a cycle $[h]$ (its link, a separating 3-sphere; on the dual 1-skeleton, the closed BFS path `wilson_line` already walks) whose period $w_h := \oint_{[h]} \psi$ is the carried weight. The color singlet is

$$(w_A, w_B, w_C) = (1, \omega, \omega^2), \quad \omega = e^{2\pi i/3}, \quad w_A + w_B + w_C = 0, \quad (1)$$

the discrete ϵ_{abc} [SR §4, §6]; the singlet $[1, 1, 1]$ is *not* carried (residual ~ 100), which is confinement.

Additivity. A cycle γ_{ij} enclosing *both* holes i and j bounds a region with those two punctures, hence is homologous to the sum of the two link cycles, and periods are linear over cycles:

$$[\gamma_{ij}] = [i] + [j], \quad \oint_{\gamma_{ij}} \psi = w_i + w_j. \quad (2)$$

This is the incidence (boundary) map of the triangle K_3 on $\{A, B, C\}$ sending each *edge* to the sum of its two endpoint *nodes*.

The duality. Impose the singlet (1) in (2):

$$\boxed{[\gamma_{AB}] = -[C], \quad [\gamma_{AC}] = -[B], \quad [\gamma_{BC}] = -[A].} \quad (3)$$

The loop around a pair is minus the complementary hole. Concretely the three pair-loops carry $(w_{AB}, w_{AC}, w_{BC}) = (-\omega^2, -\omega, -1)$, which is the *same* register $[1, \omega, \omega^2]$ up to an overall sign and a relabeling. Two immediate consequences:

- **Color is unchanged.** The loop basis carries the identical singlet, dual-labeled. So for *color* the reframe is a relabeling and adds nothing — a consistency check, not new physics. (Good: any framework in which “loops as quarks” broke the carried singlet would be wrong.)
- **No new degrees of freedom.** With $\sum_h [h] = 0$ from the singlet and the dual relation $\sum_{i<j} [\gamma_{ij}] = 2 \sum_h [h] = 0$, the three pair-loops span the *same* two-dimensional space as the three holes (mod the singlet), with one linear relation each. As S_3 representations both the node space and the edge space are $\mathbf{1} \oplus \mathbf{2}$ (trivial $\oplus$ standard): identical content, different basis.

Remark 1 (The minus sign is the antitriplet). *The conjugation $w_{ij} = -w_k$ in (3) is not cosmetic: it is the statement that the pair transforms as the conjugate of the complementary quark. This is exactly the $\bar{\mathbf{3}}$ (antitriplet) diquark of §5, seen here homologically before any group theory.*

3 Loops as quarks is the edge basis — and the spin payoff reduces to the siblings

By §2 “treat the loops as the quarks” means: change from the *node* basis $\{A, B, C\}$ to the *edge* basis $\{\gamma_{AB}, \gamma_{BC}, \gamma_{CA}\}$ of the same register. Why bother, if it adds no degrees of freedom? Because the natural *observables* differ:

- the node basis invites *per-hole* (diagonal) reads — one spinor per hole, **kron’d** — which is exactly `composite_j2` and *provably* the product floor [SR §8];
- the edge basis invites *pairwise* reads, and $J^2 = \frac{3}{4} + \sum_{i<j} P_{ij}$ is a sum over pairs [SR §3]. The edge basis is the one in which the off-diagonal swap structure of J^2 is diagonal.

This is the right instinct, but it is the *same* instinct the siblings already act on, and the honest accounting is that the *spin* content of a pair-loop reduces to objects they define. There are two distinct loop observables, and only one carries spin:

1. **The period** $\oint_{\gamma_{ij}} \psi = w_i + w_j$ ((2)). This is *linear* in ψ — a one-body datum in the internal color/taste space. It gives the duality (3) and the flavor of §4, but *not* the spin correlation, which is bilinear.
2. **The closed-loop holonomy** — the $\text{Spin}(4)$ rotation accumulated transporting a spinor around γ_{ij} . This *is* the bilocal, cross-hole object. But it is precisely the closed-loop holonomy [CW §6] names as the entangling carrier (its “escape hatch”): the pair-encircling loop γ_{ij} is the exchange loop $\Gamma_{ij} \cup \Gamma_{ji}$ of the swap $\hat{P}_{ij}$ in the swap form $J^2 = \frac{3}{4} + \sum_{i<j} P_{ij}$ [CW §3, Eq. (j2swap)], and its spinor lift is the hinge holonomy $W_{\text{spin}}(h) = \cos(\varepsilon_h/2)$ of `dk.spin.wilson.py` (the #477 spin-holonomy thread).

Claim 1 (The spin reframe is a renaming of [CW]). *Reading a spinor on the pair-loop, done correctly (the holonomy, not the period), is the closed-loop-holonomy swap $\hat{P}_{ij}$ of [CW §6] (the escape hatch); summing the three gives $\hat{J}^2 = \frac{3}{4} + \sum_{i<j} \hat{P}_{ij}$. Equivalently, the connected pairwise correlator $C_{ij} = \langle \mathbf{S}_i \mathbf{S}_j \rangle - \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_j \rangle$ of [CW §5] is the chamber coordinate of the same loop. The edge basis gives these a geometric name (“the loop around the pair”); it does not give a new mechanism.*

So the value of “loops as quarks” is *not* on the spin side, where it reduces to known plans. It is on the *flavor* and *color* sides, which the siblings treat only glancingly, and which the edge basis settles cleanly. The rest of this note is those two.

4 Flavor from the diquark loop — and why it is not an inversion

The construction. The proton is *built*, not posited [SR §7], in two steps [#489]: $q+q \rightarrow$ diquark, then diquark + $q \rightarrow$ proton. Let holes 1,2 be the two quarks that bond in step 1 (the diquark constituents) and hole 3 the quark added in step 2. The user’s proposal assigns flavor to the three loops:

$$\gamma_{12} \text{ (the diquark loop)} = |d\rangle, \quad \gamma_{13} = |u\rangle, \quad \gamma_{23} = |u\rangle. \quad (4)$$

The multiplicity is uud — one d , two u — the proton’s. The question is whether this is consistent or a sleight of hand, since in the standard picture the *diquark* is the like-flavor uu pair, so calling the diquark loop “ d ” looks backwards.

It is the Poincaré dual of the node flavor, and it agrees. It is not backwards. By the duality (3), $\gamma_{12} = -[3]$: the diquark loop *is* the dual of hole 3, the quark *outside* the diquark. In the standard uud build the diquark is $\{1,2\} = uu$ and the outside quark is $3 = d$. Hence

$$\gamma_{12} = -[3] = -[d\text{-hole}], \quad \gamma_{13} = -[2] = -[u\text{-hole}], \quad \gamma_{23} = -[1] = -[u\text{-hole}], \quad (5)$$

so the loop flavor $\{\gamma_{12}, \gamma_{13}, \gamma_{23}\} = \{d, u, u\}$ is exactly the Poincaré dual of the node flavor $\{1, 2, 3\} = \{u, u, d\}$. **The diquark loop carries d precisely because it is dual to the d -quark.** No inversion, no fiat — and, per the codebase discipline, flavor is *read off* the dual basis, not imposed [cf. the removal of `relax_singlet` for hand-placing the singlet, CW §6 caveat].

The mixed-symmetry skeleton, for free. Let $\tau = (1 \leftrightarrow 2)$ be the exchange of the two diquark constituents. On the loops,

$$\tau : \quad \gamma_{12} \mapsto \gamma_{12} \text{ (fixed)}, \quad \gamma_{13} \leftrightarrow \gamma_{23} \text{ (swapped)}. \quad (6)$$

The d -loop is the τ -spectator; the two u -loops are the τ -exchanged pair. This is exactly the proton’s defining symmetry — the mixed-symmetry state $2|uud\rangle - |udu\rangle - |duu\rangle$ is symmetric under $1 \leftrightarrow 2$ with the d in the distinguished slot [SR §3–4, Eq. (su6)]. The edge basis *produces the proton’s combinatorial skeleton* from the bare topology of which two holes bonded first. Contrast the node read, whose blind per-hole symmetrization lands on the fully symmetric quartet or the product mixture [SR §4, §8]: the loop labeling supplies precisely the structure the product read destroys.

Honest limit. This is the right *skeleton*, which is *necessary* for $\frac{3}{4}$; it is not *sufficient*. The actual value depends on the pairwise spin correlations — the bilinear holonomy/ C_{ij} data of §3, which the period (4) does not contain. The loop flavor fixes the *representation type* ($[2, 1]$ mixed, the proton channel rather than the $[3]$ symmetric Δ channel); recovering the number $\frac{3}{4}$ still requires [CW]’s closed-loop holonomy. The contribution here is that the diquark build [#489] and the proton’s spin-flavor symmetry are *the same fact* seen in two bases.

5 Loops as gluons? Rep theory says diquark

The user’s alternative — “what if the loops were gluons?” — is settled by the color representation of what a loop encloses. Two facts of $SU(3)$ Clebsch–Gordan decide it.

A two-quark loop is a diquark ($\bar{3}$), not a gluon. By Gauss’s law the color flux through a surface bounding γ_{ij} equals the total enclosed charge, here two quarks:

$$\mathbf{3} \otimes \mathbf{3} = \bar{\mathbf{3}} \oplus \mathbf{6} \quad (\text{antisymmetric } \bar{\mathbf{3}}, \text{ symmetric } \mathbf{6}). \quad (7)$$

A *gluon* is the adjoint of $\mathbf{3} \otimes \bar{\mathbf{3}}$:

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}, \quad (8)$$

which requires one quark *and one antiquark*. A loop around two *quark* holes can therefore never be in the gluon octet; it is a **diquark**. The singlet selects the antitriplet part: $\bar{\mathbf{3}} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \supset \mathbf{1}$, so the diquark loop must be the “good” $\bar{\mathbf{3}}$ to neutralize against the third quark ($\mathbf{3}$). This is the group-theoretic shadow of §2: the homological signature $[\gamma_{ij}] = -[k]$ (a sign/conjugation) *is* the $\bar{\mathbf{3}}$ being the conjugate of the complementary $\mathbf{3}$. Two derivations, Gauss law and Poincaré duality, one conclusion.

So the within-proton loops corroborate the diquark build, not a gluon picture. Far from a competing reading, “the loops are diquarks” is the same object as the $\bar{\mathbf{3}}$ diquark of the two-step build [#489] and the diquark sequence caveat [CW §7]. The edge basis makes the diquark a *first-class geometric object* (an enclosing loop), which is conceptually cleaner than tracking it as interior structure.

Where a genuine gluon loop lives. A gluon loop (8) needs a quark *and* an antiquark inside it — and those exist in this construction. The build is a simultaneous pair-creation event [SR §7], 3 neutral $q\bar{q}$ pairs $\rightarrow$ [proton, antiproton], so antiquark holes are present (each pair’s $\bar{q}$, and the antiproton leg). A loop enclosing a proton quark hole and a nearby antiquark hole is in $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$: the *meson/gluon channel*. So the correct statement is:

within-proton pair loop $\in \mathbf{3} \otimes \mathbf{3} = \bar{\mathbf{3}} \oplus \mathbf{6}$ (diquark); quark-antiquark loop $\in \mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$ (gluon/meson).

(9)

“Loops as gluons” is right *only* for the cross-leg quark-antiquark loops, not for the within-proton pair loops the diquark proposal uses.

What a gluon loop would carry — and the constituent caveat. A gluon is a spin-1 color octet: a quark-antiquark loop carries the *chromo-flux/binding* (the Y -shaped baryon string between the three color charges), contributing to mass and orbital angular momentum, not to the valence spin- $\frac{1}{2}$ content. This dovetails with [CW §4], which finds the entangling exponential acting

as the one-gluon-exchange *hyperfine* (color-magnetic spin-spin) interaction — the two views are complementary: [CW] reads the gluon as the spin-spin *operator* $\mathbf{S}_i \cdot \mathbf{S}_j$ whose sign splits N from Δ , while this note reads the gluon as the *color rep* of a $q\bar{q}$ loop. Both decline to put the proton’s $\frac{3}{4}$ on the gluon. *Aside, flagged as beyond scope:* in real QCD gluon spin and orbital angular momentum *do* carry a large part of the nucleon spin (the “spin crisis”: valence quarks supply only $\sim 30\%$) [4]. The $q\bar{q}$ loops are exactly where such contributions would live — but [SR]’s target is the constituent-model $\frac{3}{4}$ from three valence spins, and within that target the gluon loops are the glue, not the valence spin.

6 Caveats, reduction to the open kernel, and one cheap new test

1. **The edge basis is not a free lunch on spin.** By §3 the spin content of a pair-loop is the closed-loop holonomy / the correlator C_{ij} [CW §5–6]. Reading it still requires the faithful joint two-hole state, i.e. the same fiber $\leftrightarrow$ cells (Whitney / Kähler–Atiyah) lift named open in [SR §9] and [CW §6]. This note does *not* bypass that kernel; it reframes the objects feeding it.
2. **Period $\neq$ holonomy.** The flavor result §4 uses the *linear* period; the spin result needs the *bilinear* holonomy. Conflating them rebuilds the product floor in loop coordinates (the linear sum of two Bloch vectors is still a one-body object). Keep the two readings of a “loop” distinct.
3. **Three-body, not two.** A single pair-loop is a two-body object; the proton is three. As in [CW §7] the clean realization is the *diquark sequence* (bond $\{1, 2\}$, then couple 3), which is what the two-step build already does — and which the edge basis names as “form γ_{12} , then close the remaining loops.” There is no single three-body “super-loop” that linearizes the GHZ/ W structure.

The genuinely new, cheap experiment. Everything *spin* here reduces to existing plans, but one prediction of the *flavor* result is new and cheap to falsify. The dual-flavor claim (4)–(6) says: read the DK taste/charge not per hole but on the *three pair-loops* (the dual basis), and you should recover $\{d, u, u\}$ with the diquark loop γ_{12} *distinguished* as the τ -spectator. Concretely, on the built proton [#489]:

1. For each pair $\{i, j\}$, form the dual-basis period $w_i + w_j = \oint_{\gamma_{ij}} \psi$ from the already-computed per-hole weights (no new geometry), and read the DK taste on that combination.
2. Check (a) the multiplicity is 2:1 ($u:u:d$), (b) the *odd-one-out* loop coincides with the diquark loop γ_{12} (equivalently is dual to the step-2 quark hole 3), and (c) the result is invariant under the GAUGE/RELABEL gates [SR §7]. Failure of (b) would falsify the dual-flavor framing immediately; success would establish that flavor, like the diquark build, is a Poincaré-dual statement on the register — the one claim of this note that is both new and testable without the open lift.

This is a sibling of, not a substitute for, [CW]’s pairwise- C_{ij} program: that one targets *spin* (and needs the lift); this one targets *flavor* (and does not).

7 Status and references

Status. A lens, not a validated result. Established here: the pair-loop $\rightarrow$ complementary-hole duality (3) (homology) and the diquark-not-gluon classification (7)–(8) (rep theory), together with

the consistent dual-flavor labeling §4. Reframed, not resolved: the spin readout (§3 reduces it to [CW]). New and testable: the dual-basis flavor read §6.

One sentence. “Make the loops the quarks” is the Poincaré–dual edge basis of the same three color holes — it carries the identical singlet, assigns d to the diquark loop precisely because that loop is dual to the d -quark, and (by $\mathbf{3} \otimes \mathbf{3} = \bar{\mathbf{3}} \oplus \mathbf{6}$) is a diquark rather than a gluon; it sharpens flavor and color but inherits, for spin, the same closed-loop curvature and the same open lift the sibling notes already name.

Code and companions.

- Sibling documents (this directory): `spin_readout.tex` [SR] (the study document, which names the total-space operator); `cartan_weyl_gluon.tex` [CW] (the Cartan/Weyl lens, C_{ij} , the hyperfine gluon, and the fiber $\leftrightarrow$ cells lift with the closed-loop holonomy escape hatch).
- Modules: `examples/cobordism/dk_composite_spin.py` (wilson_line, facet_transport, the carried weights w_h , the product floor); `examples/cobordism/dk_spin_wilson.py` (the hinge-loop curvature); `dk_joint_spin.py`, `test_dk_joint_spin.py` (the validated J^2 instrument).
- Build/findings: `docs/design/joint_proton_spin_findings.md` (#489, the two-step diquark build and the validated measuring stick).
- Issues: #410 (flavor/charge track); #489 (build); #495 / #477 (the readout this companion’s spin reduces to).

References

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- [3] A. Hatcher, *Algebraic Topology*, Cambridge Univ. Press (2002) — homology, Poincaré duality, additivity of periods over cycles.
- [4] C. A. Aidala, S. D. Bass, D. Hartline (Hatta), and Yu. V. Kovchegov, *The spin structure of the nucleon*, Rev. Mod. Phys. **85**, 655 (2013) — the gluon spin / orbital angular momentum contributions (the “spin crisis”).
- [5] Tessera theory notes (this directory): *From Spin- $\frac{1}{2}$ to the Entangled Readout of the Proton Spin* (`spin_readout.tex`); *A Cartan / Weyl–Chamber Reading of the Composite–Spin Readout* (`cartan_weyl_gluon.tex`).